On combining (2) and (4), and remembering that the $\mathbf{a}^i$ are gradients of the $p_i(\mathbf{x})$ and that $C_1(\mathbf{x}_0) = \cdots = C_l(\mathbf{x}_0) = 0$, we have (at $\mathbf{x}_0$)

$$\nabla[\mu_1 p_1(\mathbf{x}) + \cdots + \mu_m p_m(\mathbf{x}) + \lambda_1 C_1(\mathbf{x}) + \cdots + \lambda_l C_l(\mathbf{x})] = 0, \quad (5)$$

which shows that the μ's and λ's can be interpreted as Lagrangian multipliers. This equation is closely related to a result concerning efficient points given in a paper by Kuhn and Tucker [4].

Suppose that by using the techniques sketched above or some other means[5] we have found the set Γ of noninferior systems in C. A system designer at this point would be faced with the problem of how to choose a single system from the class Γ of systems which are either not comparable or are equivalent to one another. This can be done by introducing a scalar-valued criterion which is defined on Γ but not necessarily C, or, alternatively, tossing a die. In the opinion of the writer, a two-stage process which involves as a first step the determination of the set Γ of noninferior systems and then the choosing of a single system in Γ makes more sense than the conventional approach in which a scalar-valued criterion defined on Σ is introduced at the outset.

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[4] H. W. Kuhn and A. W. Tucker, "Nonlinear programming," *Proc. Second Berkeley Symp. on Math. Stat. and Prob.*, Berkeley, Calif., University of California Press; 1951.

[5] Dynamic programming, which offers an effective means of finding the set of noninferior points in the case of finite-state systems, is one example.

Minimal Effort Control Systems with an Upper Bound on the Control Time*

I. Introduction

This note is concerned with a linear control system described by

$$\frac{dx}{dt} = A(t)x(t) + B(t)u(t)$$

where a set of admissible control variables Ω and a control effort functional $\mathcal{E}(u)$ are given. Here,

$$\mathcal{E}(u) = \int_0^t \phi(u(\tau))d\tau$$

and $A(t)$ is an $n \times n$ matrix, $x(t)$ is an n vector, $B(t)$ is an $n \times r$ matrix and $u(t)$ is an r vector.

In the past the above system has been considered by various authors to bring the system from its initial state x_0 to a state designated by x_1 minimizing either

1) $\mathcal{E}(u)$,

or

2) the time t, to reach x_1 where $x_1 = x(t_1)$ and $x_0 = x(0)$.

For historical development see, for example, Neustadt.[1] In his paper, he requires among other things that the function $\phi(u)$ be convex. It is essential for his computational algorithm[2] that the set $C(t)$ of points in the state space from which the desired final state can be reached in time t to be closed, convex, and that ϕ satisfies what he terms as "the generalized unique maximum condition."

It is clear that the condition he imposes on $\phi(u)$ is sufficient but not necessary.[3] He considers minimizing control effort for the specific control period T.

It is more realistic to consider the problem of minimizing control effort subject to the upper bound on the available control time or the problem of minimizing the time to reach the desired state with a bounded available control effort to accomplish the task.

This note discusses the above-mentioned problem 1) for a simple example where $\phi(u)$ is taken to be nonconvex.

II. A Simple Example

A. Introduction

Consider the first-order system,

$$\frac{dx}{dt} = -ax + bu \qquad a, b > 0$$

$$\mathcal{E}(u) = \int_0^t \ln(1 + |u(\tau)|)d\tau$$

$$\Omega_1 = \{u = 1, u = -1\}$$

$$\Omega_2 = \{u; |u| \leq 1, u \text{ measurable}\}$$

and

$$\Omega_3 = \{u = -1, u = 0, u = +1\}.$$

Let

$$x_1(t) = \int_0^t \ln(1 + |u(\tau)|)d\tau, \qquad x_1(0) = 0$$

$$x_2(t) = \int_0^t e^{as}bu(s)ds$$

$$C(t) = \{x_1(t), x_2(t); u \text{ admissible}\}.$$

Then, the problem of reaching the origin with minimizing $\mathcal{E}(u)$ has the geometric interpretation of finding a point with a minimal x_1 coordinate among those $C(t)$ which intersects the line $x_2 = -x(0)$, the set of desired final points, E.

Note that

$$C(t_1) \subset C(t_2), \qquad t_1 \leq t_2 \quad \text{for } \Omega_2$$

so long as $\phi(u(\tau)) \geq 0$.

Since E is a cylinder set with respect to x_1 coordinate, the point with minimal x_1 coordinate in the set $C(t) \cap E$ will be on the boundary of $C(t)$.[4]

If $\xi \in E$, if ξ is on the boundary of $C(t_1)$ and if $x_2(\tau)$, $0 \leq \tau \leq t_1$ is a trajectory with $x_2(t_1) = \xi$ using an admissible control, then the whole trajectory $x_2(\tau)$ belongs to the boundary of $C(\tau)$, $0 \leq \tau \leq t_2$. The proof can be supplied by the adaptation of the proof in Theorem 3.1 of Roxin.[4]

Thus, it becomes essential to obtain the boundary of $C(t)$. The set $C(t)$ is symmetric with respect to the x_1 axis due to the choice of $\phi(u)$.

B. Set Ω_1

On the set Ω_1,

$$|u| = 1 \quad \text{a.e.}$$

Hence,

$$x_1 = \int_0^t \ln(1 + (u))d\tau = (\ln 2)$$

$$x_2 \leq \int_0^t e^{as}bu(s)ds \leq \frac{b}{a}(e^{at} - 1).$$

The $C(t)$ is shown in Figs. 1 and 2.

Therefore, one sees from Fig. 2 that the system can reach the origin in time t_2 with $\mathcal{E}(u) = (\ln 2)t_2$ and that so long as $t_2 \leq T$ where T is the upper bound on the total available control time, T does not impose any additional constraint on the problem and that one does not benefit from having T greater than t_2.

C. Set Ω_2

The set $C(t)$ in this case is shown in Fig. 3. The point A has the coordinate

$$x_1 = (\ln 2)t$$

$$x_2 = \frac{b}{a}(e^{at} - 1).$$

The fact that the set is convex can be seen from the following elementary considerations. Consider a set S on which $|u| = 1$, and let the measure of S be $P_1 t$.

Then

$$x_1 = \int_S \ln(1 + |u|)d\tau$$

$$= (\ln 2)P_1 t \triangleq \xi$$

$$\frac{b}{a}(e^{aP_1 t} - 1) = \int_0^{P_1 t} e^{as}b \cdot ds \leq x_2$$

$$= \int_S e^{as}bu(s)ds \leq \int_{(1-P_1)t}^t e^{as}b \cdot 1\, d\tau$$

$$= \frac{b}{a}(e^{aP_1 t} - 1)e^{a(1-P_1)t}.$$

[1] L. W. Neustadt, "Time optimal systems with position and integral limits," *J. Math. Analysis and Application*, vol. 3, pp. 406–427; December, 1961.

[2] L. W. Neustadt, "Synthesizing time optimal control systems," *J. Math. Analysis and Application*, vol. 1, pp. 484–493; December, 1960.

[3] The condition on $\phi(u)$ for the set $C(t)$ to be convex can be relaxed considerably. See, for example, H. Halkin, "Liapounov's Theorem on the Range of a Vector Measure and Pontryagin's Maximum Principle," Appl. Math. and Stat. Labs., Stanford University, Stanford, Calif., Tech. Rept. No. 109; May, 1962.

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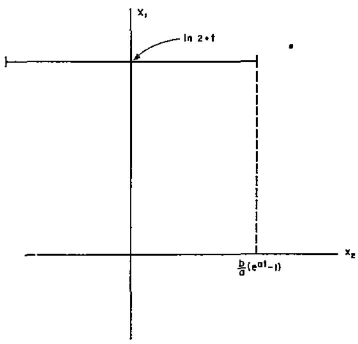

Fig. 1—Set $C(t)$ with Ω_1.

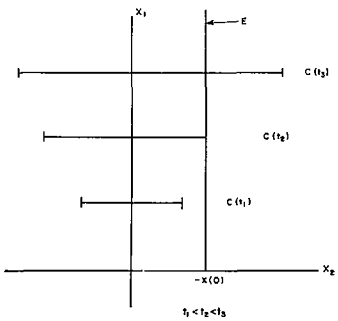

Fig. 2—Set $C(t)$ for various t with Ω_1.

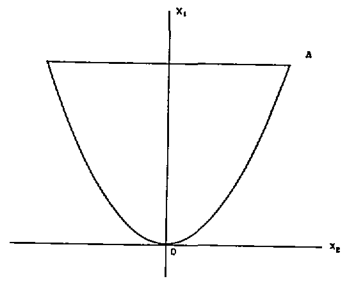

Fig. 3—Set $C(t)$ with Ω_2.

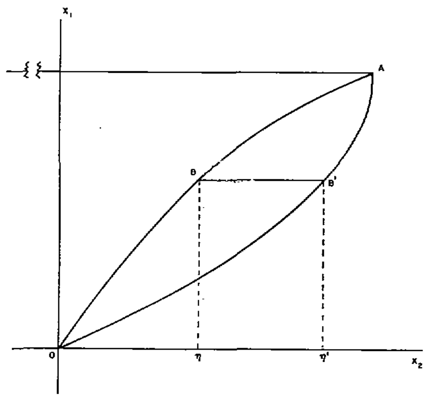

Fig. 4—Boundary of $C(t)$.

These inequalities have the interpretation that if a point B with η as its x_2 coordinate can reach the origin in time $P_1 t$ with control effort $(\ln 2)(P_1 t)$ then, by waiting during the time period from t to $P_1 t$, one can reach the orgin from the points whose x_2 coordinate is given by $\eta e^{(1-P_1)t}$.

Thus, referring to Fig. 4, the curve from 0 to B with $\eta = b/a(e^{aP_1 t}-1)$ can be reached using $u(\tau)=1, 0 \le \tau \le P_1 t$. The point B' with $\eta' = \eta e^{(1-P_1)t}$ can also reach the origin with the same control effort and since η' is the upper limit on x_2 using control variables from Ω_2 and maintaining the same control effort, the boundary of $C(t)$ is given by considering B' points at each control effort level up to $x_1 = (\ln 2)t$. The coordinates of the boundary curve A 0 are given in a parametric form as

$$x_1(\tau) = (\ln 2)\tau, \qquad 0 \le \tau \le t.$$
$$x_2(\tau) = \frac{b}{a}\left(e^{at} - e^{a(t-\tau)}\right)$$

It can be easily verified that

$$\frac{d^2 x_1}{dx_2{}^2} > 0;$$

therefore, the boundary curve of $C(t)$ is convex.

In Fig. 5, the corner of the set $C(t_i)$ is denoted by A_i. Points A_i's lie on the curve

$$x_1 = \frac{(\ln 2)}{a} \ln\left(1 + \frac{a}{b} x_2\right).$$

Here, the advantage of having a sufficiently large T is apparent. Namely, if $-x(0) \in C(t_1) \cap E$ for the first time, then the system can reach the origin from $-x(0)$ in time t_1 with the control effort $(\ln 2)t_1$.

If $t_1 < T$, then it is advantageous to make use of this additional control time and the

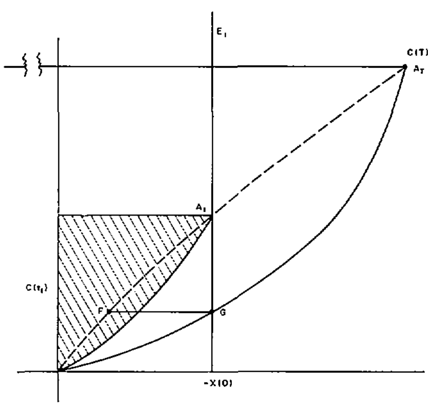

Fig. 5—Set $C(t)$ on its boundary.

system can reach the origin with the control effort less than $(\ln 2)t_1$. Physically, this has the interpretation that the system coasts for the first part of the control to the point F, then employs $|u|=1$ to reach the origin.

D. Set Ω_3

Note that the set $C(t)$ for Ω_3 is identical with that for Ω_2.

Hence, in mechanizing the system one needs only to construct an actuator with three positions -1, 0 and $+1$.[5]

III. Closing Comments

As a final comment, in the time optimal problems with control effort constraint, if the available control effort is larger than $(\ln 2)t_1$ where $x_2(t_1) = x_2$, then this does not constitute any real constraint.

It is clear from the above example that whenever the set $C(t)$ satisfies the condition $C(t_1) \subset C(t_2)$, $t_1 < t_2$, the control effort will be reduced by allowing more time for control until the longest allowed control time T is reached. The extreme example of this is given by a stable dynamic system which returns to the origin with no control effort when T is made infinite. This has the geometric interpretation that the boundary of $C(\infty)$ contains x_2 axis.

IV. Acknowledgment

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On Stability and "Steepest Descent"*

Recently a number of adaptive optimizing schemes were proposed [1]–[3] in which the controllable parameters are continuously adjusted along a path of "steepest descent" on the objective function vs parameters surface toward an optimum point (possibly moving). The success of these methods depends, of course, on the objective function having a unique minimum (or maximum) point and, further, that the minimum (maximum) be a minimum (maximum) in the calculus sense. It is the purpose of this communication to examine the stability of an idealized version of these schemes using Lyapunov's second method [4] and to isolate the causes of nonoptimum adjustment.

Let $y_1, y_2, \cdots, y_n$ denote the actual (adjustable) parameters of the system, $z_1, z_2, \cdots, z_n$ be the corresponding optimum parameter values and $\boldsymbol{x} = \boldsymbol{y} - \boldsymbol{z}$ represent the (vector) error between the actual and optimum parameter values. Consider a scalar objective function, $V(\boldsymbol{x}; t)$ which has a unique minimum at $\boldsymbol{x} = 0$, $V(0; t) = 0$, has continuous partial derivatives and which is positive definite. Note that for $V(\boldsymbol{x}; t)$ to be positive definite there must exist a $W(\boldsymbol{x})$ which is positive definite and $0 < W(\boldsymbol{x}) \le V(\boldsymbol{x}; t)$ for $\boldsymbol{x} \ne 0$ and for all t in the region of consideration. For example, the objective function might be the normalized MSE or MAE in a control system. If the parameters $y_1, y_2, \cdots, y_n$ are independently adjustable and are adjusted along the path of instantaneous "steepest descent" [5], then

$$\dot{y}_j = -k_0 \frac{\partial V}{\partial y_j}, \quad k_0 > 0, \; j = 1, 2, \cdots, n. \quad (1)$$

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